

Bit rate = 4 Mbps

Since, alternate QPSK and 16-QAM used, data rate of uncoded data is

$$= \frac{4 \text{ Mbps} + 2 \text{ Mbps}}{2} = 3 \text{ Mbps.}$$

End of Solution

Q.36 The function $f(x) = 8\log_e x - x^2 + 3$ attains its minimum over the interval $[1, e]$ at $x =$ _____.

(Here $\log_e x$ is the natural logarithm of x .)

- (a) 2 (b) 1
(c) e (d) $\frac{1+e}{2}$

Ans. (b)

$$f(x) = 8\ln(x) - x^2 + 3$$

Stationary points,

$$f'(x) = \frac{8}{x} - 2x = 0$$

$$2x^2 = 8$$

$$x = \pm 2$$

$$x = 2 \in [1, e]$$

$$f''(x) = -\frac{8}{x^2} - 2 \Big|_{\text{at } x=2} = -\frac{8}{4} - 2 = -6$$

$$f''(2) < 0$$

$\therefore f(x)$ is maximum.

$\therefore$ Minimum of $f(x)$ in $[1, e]$

$$= \min \{f(1), f(e)\}$$

$$= \min \{2, f(e)\}$$

$$f(e) = 8\ln e - e^2 + 3$$

$$= 8 + 3 - e^2$$

$$= 11 - 7.38$$

$$f(e) = 3.61$$

$\therefore$ Minimum value of $f(x)$ occurs at $x = 1$.

End of Solution

Q.37 Let α, β be two non-zero real numbers and v_1, v_2 be two non-zero real vectors of size 3×1 . Suppose that v_1 and v_2 satisfy $v_1^T v_2 = 0, v_1^T v_1 = 1$, and $v_2^T v_2 = 1$. Let A be the 3×3 matrix given by :

$$A = \alpha v_1 v_1^T + \beta v_2 v_2^T$$

The eigen values of A are _____.

- (a) 0, α, β (b) 0, $\alpha + \beta, \alpha - \beta$
(c) 0, $\frac{\alpha + \beta}{2}, \sqrt{\alpha\beta}$ (d) 0, 0, $\sqrt{\alpha^2 + \beta^2}$

Ans. (a)

Given, $A = \alpha v_1 v_1^T + \beta v_2 v_2^T$

Post multiply by v_1 on both sides,

We get, $Av_1 = (\alpha v_1 v_1^T + \beta v_2 v_2^T)v_1$

$$Av_1 = \alpha v_1 v_1^T v_1 + \beta v_2 v_2^T v_1$$

$$Av_1 = \alpha v_1$$

$\Rightarrow \alpha$ is an eigen value of A .

Post multiply by v_2 on both sides,

We get $Av_2 = \alpha v_1 v_1^T v_2 + \beta v_2 v_2^T v_2$

$$Av_2 = \beta v_2$$

$\Rightarrow \beta$ is an eigen value of A .

$v_1 v_1^T$ and $v_2 v_2^T$ both are singular matrices.

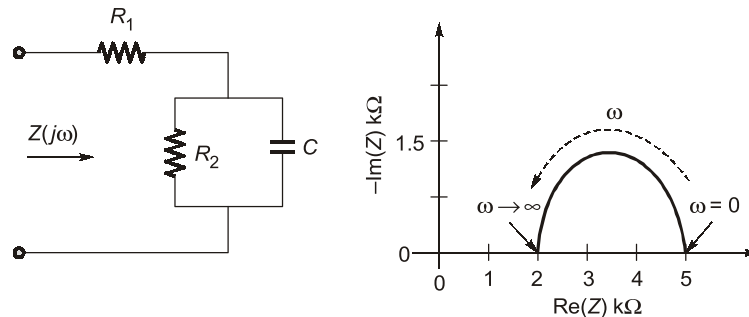
$\therefore A$ is also a singular matrix

$$\Rightarrow |A| = 0 \Rightarrow \lambda_A = 0$$

Hence, $\lambda_A = 0, \alpha, \beta$

End of Solution

Q.38 For the circuit shown, the locus of the impedance $Z(j\omega)$ is plotted as ω increases from zero to infinity. The values of R_1 and R_2 are:



(a) $R_1 = 2 \text{ k}\Omega, R_2 = 3 \text{ k}\Omega$

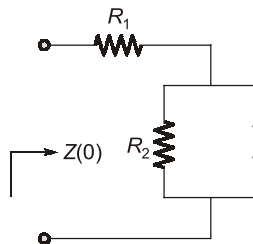
(b) $R_1 = 5 \text{ k}\Omega, R_2 = 2 \text{ k}\Omega$

(c) $R_1 = 5 \text{ k}\Omega, R_2 = 2.5 \text{ k}\Omega$

(d) $R_1 = 2 \text{ k}\Omega, R_2 = 5 \text{ k}\Omega$

Ans. (a)

At $\omega = 0 \text{ rad/s}$, $X_C = \frac{1}{\omega C} = \infty$



Impedance, $Z(0) = R_1 + R_2$

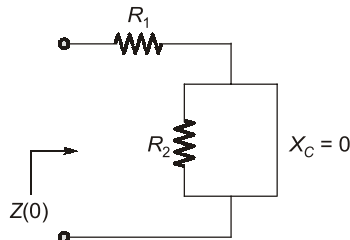
With the help of graph at $\omega = 0$.

$$Z(0) = 5 \text{ k}\Omega$$

$$R_1 + R_2 = 5 \text{ k}\Omega$$

At $\omega = \infty \text{ rad/s}$, $X_C = \frac{1}{\omega C}$

$$X_C = 0$$



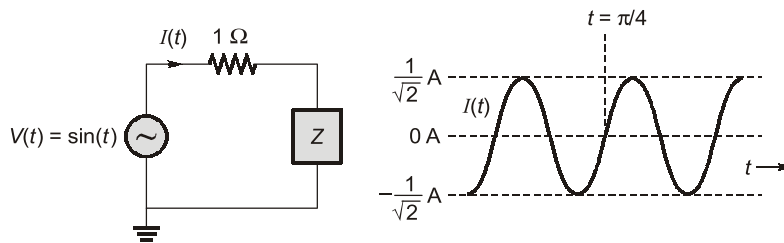
Impedance, $Z(\infty) = R_1 = 2 \text{ k}\Omega$

$$R_1 + R_2 = 5 \text{ k}\Omega$$

$$R_2 = 3 \text{ k}\Omega$$

End of Solution

- Q.39** Consider the circuit shown in the figure with input $V(t)$ in volts. The sinusoidal steady state current $I(t)$ flowing through the circuit is shown graphically (where t is in seconds). The circuit element Z can be _____.



- (a) a capacitor of 1 F (b) an inductor of 1 H
(c) a capacitor of $\sqrt{3} \text{ F}$ (d) an inductor of $\sqrt{3} \text{ H}$

Ans. (b)

As the current $i(t)$ is lagging, element Z is inductor,

$$I = \frac{V}{Z_0}$$

$$I = \frac{\sin t}{Z_0}$$

Maximum value of current $i(t) = \frac{1}{\sqrt{2}}$

$\therefore$

$$|Z_0| = \sqrt{2}$$

$$Z_0 = R + jX_L = 1 + j\omega L$$



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$$\frac{\sqrt{1 + \omega^2 L^2}}{1 + \omega^2 L^2} = \frac{\sqrt{2}}{2}$$

given,

$$\omega = 1 \text{ rad/sec}$$

$$L^2 = 1$$

$$L = 1 \text{ H}$$

End of Solution

- Q.40** Consider an ideal long channel n MOSFET (enhancement-mode) with gate length $10 \mu\text{m}$ and width $100 \mu\text{m}$. The product of electron mobility (μ_n) and oxide capacitance per unit area (C_{ox}) is $\mu_n C_{ox} = 1 \text{ mA/V}^2$. The threshold voltage of the transistor is 1 V . For a gate-to-source voltage $V_{GS} = [2 - \sin(2t)] \text{ V}$ and drain-to source voltage $V_{DS} = 1 \text{ V}$ (substrate connected to the source), the maximum value of the drain-to-source current is _____.
(a) 40 mA (b) 20 mA
(c) 15 mA (d) 5 mA

Ans. (c)

$$V_{GS} = [2 - \sin 2t] \text{ Volt}$$

$$V_{GS\min} = 2 - 1 = 1 \text{ V}$$

$$V_{GS\max} = 2 - (-1) = 3 \text{ V}$$

$$V_{DS} = 1 \text{ V}$$

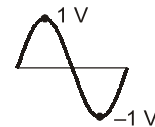
$$V_{DS} < V_{GS} - V_t \text{ (Linear)}$$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left\{ (V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right\}$$

$$I_{D\max} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS\max} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

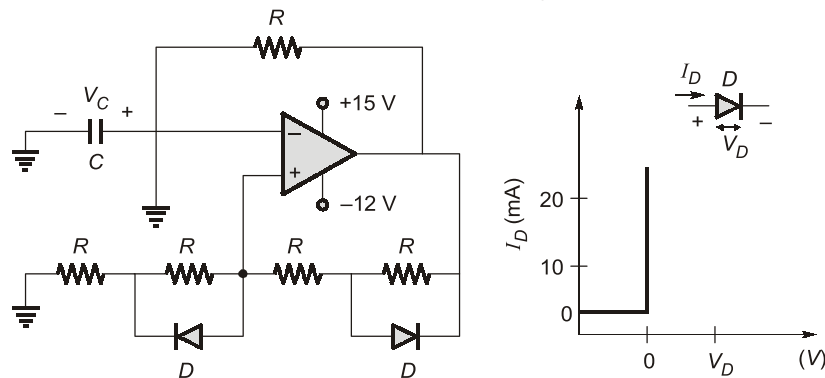
$$= 1 \text{ mA/V}^2 \times 10 \left[(3 - 1)1 - \frac{1}{2} \times 1 \right]$$

$$= 10[1.5] = 15 \text{ mA}$$



End of Solution

- Q.41** For the following circuit with an ideal OPAMP, the difference between the maximum and the minimum values of the capacitor voltage (V_C) is _____.
(a) 15 V (b) 27 V
(c) 13 V (d) 14 V



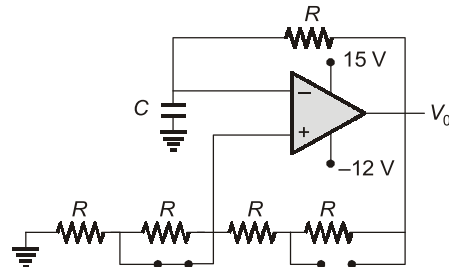
- (a) 15 V (b) 27 V
(c) 13 V (d) 14 V

Ans. (c)

If $V_0 = +15 \text{ V}$

D_1 is ON

D_2 is OFF



Capacitor charge upto V_{UT} ,

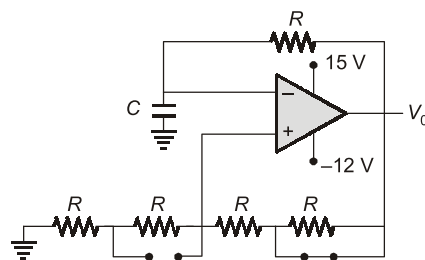
$$V_{C \max} = V_{UT} = \frac{15 \times R}{R + 2R} = 5 \text{ V}$$

If $V_0 = -12 \text{ V}$

D_1 is OFF

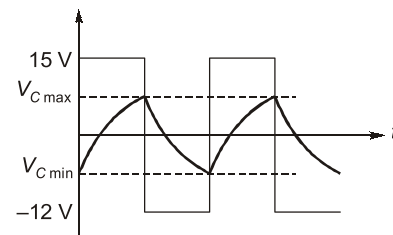
D_2 is ON

Capacitor discharges upto V_{LT} .



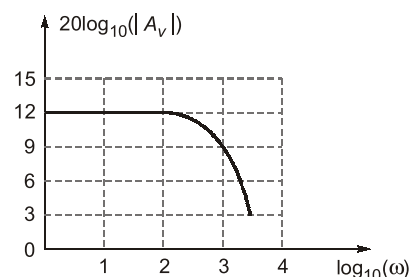
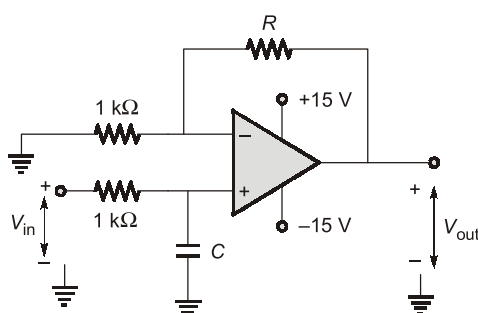
$$V_{C \min} = V_{LT} = \frac{-12 \times 2R}{2R + R} = -8 \text{ V}$$

$$V_{C \max} - V_{C \min} = 5 - (-8) = 13 \text{ V}$$



End of Solution

Q.42 A circuit with an ideal OPAMP is shown. The Bode plot for the magnitude (in dB) of the gain transfer function ($A_V(j\omega) = V_{out}(j\omega)/V_{in}(j\omega)$) of the circuit is also provided (here, ω is the angular frequency in rad/s). The values of R and C are _____.



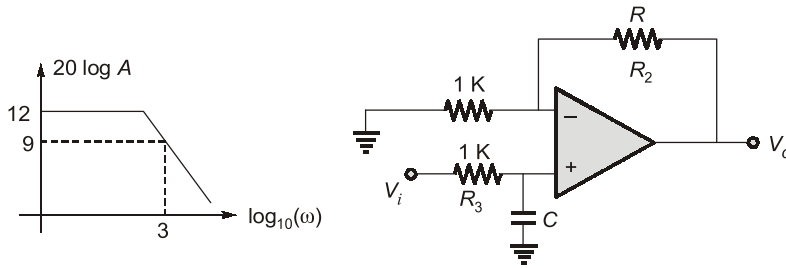
(a) $R = 3 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$

(b) $R = 1 \text{ k}\Omega$, $C = 3 \text{ }\mu\text{F}$

(c) $R = 4 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$

(d) $R = 3 \text{ k}\Omega$, $C = 2 \text{ }\mu\text{F}$

Ans. (a)



Maximum gain = 12 dB

$$20 \times \log A_{\max} = 12$$

$$A_{\max} = 4$$

$$1 + \frac{R_2}{R_1} = 4 \Rightarrow R_2 = 3R_1$$

$\Rightarrow$

$$R = 3 \times 1 = 3 \text{ k}\Omega$$

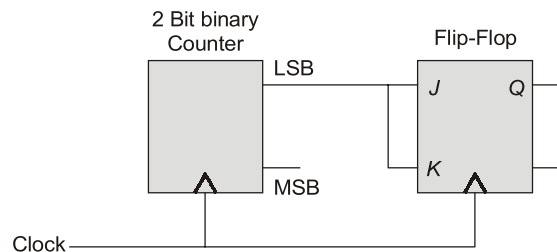
$$\log_{10} \omega_c = 3 \Rightarrow \omega_c = 1000 \text{ rad/sec}$$

$$\omega_c = \frac{1}{R_3 C} \Rightarrow C = \frac{1}{R_3 \times \omega_c}$$

$$C = \frac{1}{1000 \times 1000} = 1 \mu\text{F}$$

End of Solution

Q.43 For the circuit shown, the clock frequency is f_0 and the duty cycle is 25%. For the signal at the Q output of the Flip-Flop, _____.



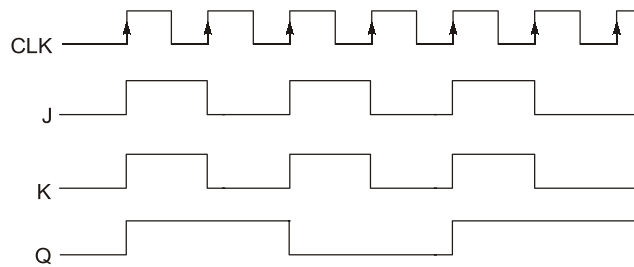
- (a) frequency is $f_0/4$ and duty cycle is 50%
- (b) frequency is $f_0/4$ and duty cycle is 25%
- (c) frequency is $f_0/2$ and duty cycle is 50%
- (d) frequency is f_0 and duty cycle is 25%

Ans. (a)

2 bit counter

MSB LSB(J, K)

0	0
0	1
1	0
1	1



$\therefore$ Duty cycle = 50%

$$\text{Output frequency} = \frac{f_o}{4}$$

End of Solution

Q.44 Consider an even polynomial $p(s)$ given by

$$p(s) = s^4 + 5s^2 + 4 + K$$

where K is an unknown real parameter. The complete range of K for which $p(s)$ has all its roots on the imaginary axis is _____.

- (a) $-4 \leq K \leq \frac{9}{4}$ (b) $-3 \leq K \leq \frac{9}{2}$
(c) $-6 \leq K \leq \frac{5}{4}$ (d) $-5 \leq K \leq 0$

Ans. (a)

Given:

$$p(s) = s^4 + 5s^2 + (4 + K)$$

Routh's Table:

$$\begin{array}{c|ccc} s^4 & 1 & 5 & (4+K) \\ s^3 & 0 & 0 & \\ s^2 & & & \\ s^1 & & & \\ s^0 & & & \end{array}$$

As all row of s^3 are zero,

$$A(s) = s^4 + 5s^2 + (4 + K)$$

$$\frac{dA(s)}{ds} = 4s^3 + 10s + 0$$

$$\begin{array}{c|ccc} s^4 & 1 & 5 & (4+K) \\ s^3 & 4 & 10 & 0 \\ s^2 & \frac{10}{4} - \frac{5}{2} & (4+K) & \\ s^1 & \frac{25 - 4(4+K)}{5/2} & & \\ s^0 & 4+K & & \end{array}$$

1st column of R-H table must be all positive.

$$\text{i.e., } \frac{25 - 4(4 + K)}{5/2} > 0 \Rightarrow K < \frac{9}{4}$$

$$4 + K > 0 \Rightarrow K > -4$$

$$\text{Range of } K: -4 < K < \frac{9}{4}$$

End of Solution

Q.45 Consider the following series:

$$\sum_{n=1}^{\infty} \frac{n^d}{c^n}$$

For which of the following combinations of c, d values does this series converge?

- (a) $c = 1, d = -1$ (b) $c = 2, d = 1$
(c) $c = 0.5, d = -10$ (d) $c = 1, d = -2$

Ans. (b, d)

$$c = 2, d = 1$$

$$\sum u_n = \sum \frac{n}{2^n}$$

$$\text{Ratio test : } \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{n+1}{2^{n+1}} \times \frac{2^n}{n} = \frac{1}{2}$$

$$\frac{1}{2} < 2$$

$\therefore$ By ratio test, $\sum u_n$ is convergent.

(a)

$$c = 1, d = -1$$

$$\sum u_n = \sum \frac{1}{n} \text{ is divergent by } P\text{-test}$$

(c)

$$c = 0.5, d = -10$$

$$\sum u_n = \sum \frac{n^{-10}}{(0.5)^n}$$

$$\text{Ratio test : } \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^{-10}}{(0.5)^{n+1}} \times \frac{(0.5)^n}{n^{10}}$$

$$= \frac{1}{0.5} = 2$$

$$2 > 1$$

$\sum u_n$ is divergent.

(d)

$$c = 1, d = -2$$

$$\sum u_n = \sum \frac{n^{-2}}{(1)^n} = \sum \frac{1}{n^2}$$

$\sum u_n$ is convergent by P -test.

End of Solution

Q.46 The outputs of four systems (S_1 , S_2 , S_3 , and S_4) corresponding to the input signal $\sin(t)$, for all time t , are shown in the figure.

Based on the given information, which of the four systems is/are definitely NOT LTI (linear and time-invariant)?

$$\sin(t) \longrightarrow S_1 \longrightarrow \sin(-t)$$

$$\sin(t) \longrightarrow S_2 \longrightarrow \sin(t+1)$$

$$\sin(t) \longrightarrow S_3 \longrightarrow \sin(2t)$$

$$\sin(t) \longrightarrow S_4 \longrightarrow \sin^2(t)$$

(a) S_1

(b) S_2

(c) S_3

(d) S_4

Ans. (c, d)

$$\sin t \longrightarrow S_1 \longrightarrow \sin(-t) = -\sin t$$

$$\sin t \longrightarrow S_2 \longrightarrow \sin(t+1)$$

$$\sin t \longrightarrow S_3 \longrightarrow \sin(2t)$$

$$\sin t \longrightarrow S_4 \longrightarrow \sin^2(t) = \frac{1 - \cos 2t}{2}$$

Since, LTI system does not change the frequency of sinusoidal input. So S_3 and S_4 are definitely not LTI as input and output sinusoidal frequencies are different.

End of Solution

Q.47 Select the CORRECT statement(s) regarding semiconductor devices.

- (a) Electrons and holes are of equal density in an intrinsic semiconductor at equilibrium.
- (b) Collector region is generally more heavily doped than Base region in a BJT.
- (c) Total current is spatially constant in a two terminal electronic device in dark under steady state condition.
- (d) Mobility of electrons always increases with temperature in Silicon beyond 300 K.

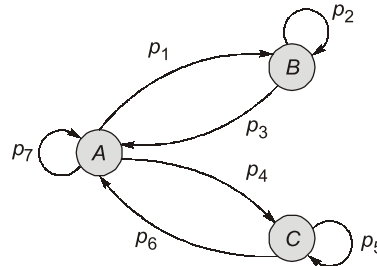
Ans. (a, c)

- In intrinsic semiconductor at equilibrium $n = p = n_i$
- Collector region is generally lightly doped than base region in a BJT.
- Total current is spatially constant in a two terminal electronic device, however individual currents vary spatially under dark and steady state condition.
- Beyond 300K, mobility of electron decreases with increases in temperature.

Hence, statement(s) : (a) and (c) are correct.

End of Solution

- Q.48** A state transition diagram with states A , B , and C , and transition probabilities $p_1, p_2, \dots, p_7$ is shown in the figure (e.g., p_1 denotes the probability of transition from state A to B). For this state diagram, select the statement(s) which is/are universally true.



- (a) $p_2 + p_3 = p_5 + p_6$ (b) $p_1 + p_3 = p_4 + p_6$
(c) $p_1 + p_4 + p_7 = 1$ (d) $p_2 + p_5 + p_7 = 1$

Ans. (a, c)

$$\left. \begin{array}{l} p_2 + p_3 = 1 \\ p_5 + p_6 = 1 \end{array} \right\} p_2 + p_3 = p_5 + p_6$$

Option (a) is correct.

$$p_1 + p_4 + p_7 = 1$$

Option (c) is correct.

End of Solution

- Q.49** Consider a Boolean gate (D) where the output Y is related to the inputs A and B as, $Y = A + \bar{B}$, where $+$ denotes logical OR operation. The Boolean inputs '0' and '1' are also available separately. Using instances of only D gates and inputs '0' and '1', _____ (select the correct option(s)).

- (a) NAND logic can be implemented (b) OR logic cannot be implemented
(c) NOR logic can be implemented (d) AND logic cannot be implemented

Ans. (a, c)

$$F(A, B) = A + \bar{B}$$

As 0 and 1 are available.

$$\begin{aligned} F(0, B) &= A + \bar{B} = 0 + \bar{B} \\ &= \bar{B} \text{ (NOT)} \end{aligned}$$

$$F(A + \bar{B}) = A + \bar{\bar{B}} = A + B$$

$$F(A + \bar{B}) = A + B \text{ (OR)}$$

With the combination of OR and NOT, NOR gate can be implemented.

Since NOR gate is universal logic gate, so all the functions can be implemented.

So, correct option is (a, c).

End of Solution

Q.50 Two linear time-invariant systems with transfer functions

$$G_1(s) = \frac{10}{s^2 + s + 1} \text{ and } G_2(s) = \frac{10}{s^2 + s\sqrt{10} + 10}$$

have unit step responses $y_1(t)$ and $y_2(t)$, respectively. Which of the following statements is/are true?

- (a) $y_1(t)$ and $y_2(t)$ have the same percentage peak overshoot.
- (b) $y_1(t)$ and $y_2(t)$ have the same steady-state value.
- (c) $y_1(t)$ and $y_2(t)$ have the same damped frequency of oscillation.
- (d) $y_1(t)$ and $y_2(t)$ have the same 2% settling time.

Ans. (a)

For system $G_1(s) = \frac{10}{s^2 + s + 1}$

Characteristics equation,

$$s^2 + s + 1 = 0$$

The standard characteristics equation is

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

On comparing, $\omega_n = 1, \xi = \frac{1}{2} = 0.5$

$$\omega_d = \omega_n \sqrt{1 - \xi^2} = 1 \sqrt{1 - (0.5)^2} = 0.866$$

Settling time, $t_s = \frac{4}{\xi\omega_n} = 8 \text{ sec}$

Steady-state error,

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \left(\frac{1}{s} \right) \cdot 10}{s^2 + s + 1} = 10$$

$$e_{ss} = 10$$

For system, $G_2(s) = \frac{10}{s^2 + \sqrt{10}s + 10}$

Characteristics equation,

$$s^2 + \sqrt{10}s + 10 = 0$$

Standard characteristics equation,

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

On comparing, $\omega_n^2 = 10 \Rightarrow \omega_n = \sqrt{10}$

$$2\xi\omega_n = \sqrt{10} \Rightarrow \xi = 0.5$$

$$\omega_d = \omega_n \sqrt{1 - \xi^2} = \sqrt{10} \sqrt{1 - (0.5)^2} = 2.739$$

Settling time, $t_s = \frac{4}{\xi\omega_n} = \frac{4}{0.5\sqrt{10}} = \frac{8}{\sqrt{10}} = 2.535$

Steady state error, $e_{ss} = \lim_{s \rightarrow 0} \frac{s \left(\frac{1}{s} \right) \cdot 10}{s^2 + \sqrt{10}s + 10} = 1$

$e_{ss} = 1$

Since 'ξ' value for both the system is same. So percentage peak overshoot for both system is same.

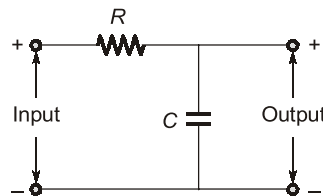
End of Solution

Q.51 Consider an FM broadcast that employs the pre-emphasis filter with frequency response

$$H_{pe}(\omega) = 1 + \frac{j\omega}{\omega_0},$$

where $\omega_0 = 10^4$ rad/sec.

For the network shown in the figure to act as a corresponding de-emphasis filter, the appropriate pair(s) of (R, C) values is/are _____.



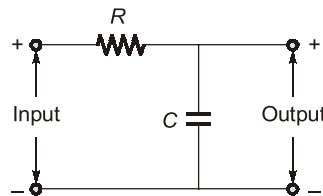
- (a) $R = 1 \text{ k}\Omega$, $C = 0.1 \text{ }\mu\text{F}$ (b) $R = 2 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$
(c) $R = 1 \text{ k}\Omega$, $C = 2 \text{ }\mu\text{F}$ (d) $R = 2 \text{ k}\Omega$, $C = 0.5 \text{ }\mu\text{F}$

Ans. (a)

Given frequency response of pre-emphasis filter,

$$H_{pe}(\omega) = 1 + \frac{j\omega}{\omega_0} \text{ where, } \omega_0 = 10^4 \frac{\text{rad}}{\text{sec}}$$

Deemphasis filter given as



$$H_{De}(\omega) = \frac{1}{1 + j\omega RC} \longrightarrow |H_{De}(\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

Relationship between pre-emphasis and De-emphasis systems is

$$|H_{pe}(\omega)| = \frac{1}{|H_{De}(\omega)|}$$

$$|H_{Pe}(\omega)|^2 = \frac{1}{|H_{De}(\omega)|^2}$$

$$1 + \left(\frac{\omega}{\omega_o}\right)^2 = 1 + (\omega RC)^2$$

$$\frac{1}{\omega_o} = RC \Rightarrow RC = 10^{-4}$$

Option (a) only satisfies the required condition.

End of Solution

Q.52 A waveguide consists of two infinite parallel plates (perfect conductors) at a separation of 10^{-4} cm, with air as the dielectric. Assume the speed of light in air to be 3×10^8 m/s. The frequency/frequencies of TM waves which can propagate in this waveguide is/are _____.

- (a) 6×10^{15} Hz (b) 0.5×10^{12} Hz
(c) 8×10^{14} Hz (d) 1×10^{13} Hz

Ans. (a, c)

The TM mode that propagates at the lowest cut-off frequency is TM_1 mode. Hence,

$$f_c|_{TM_1} = \frac{mc}{2a} ; m=1$$

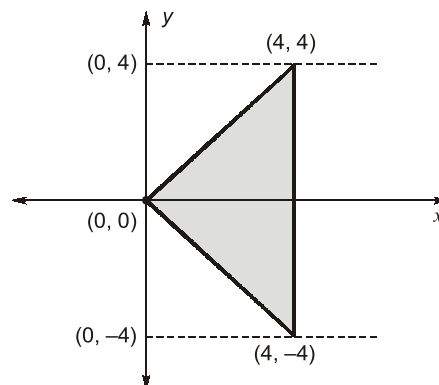
$$\therefore f_c = \frac{3 \times 10^{10}}{2 \times 10^{-4}} = 1.5 \times 10^{14}$$

End of Solution

Q.53 The value of the integral

$$\iint_D 3(x^2 + y^2) dx dy,$$

where D is the shaded triangular region shown in the diagram, is _____ (rounded off to the nearest integer).



Ans. (512)

$$\begin{aligned}
 I &= \int_{0-x}^4 \int_{-x}^x (3x^2 + 3y^2) dy dx \\
 &= \int_0^4 \left(3x^2 y + \frac{3y^3}{3} \right)_{-x}^x dx = \int_0^4 (3x^2(2x) + 2x^3) dx \\
 &= \int_0^4 (6x^3 + 2x^3) dx = \int_0^4 8x^3 dx \\
 &= \frac{8}{4} (x^4)_0^4 = 2 \times 4^4 = 512
 \end{aligned}$$

End of Solution

- Q.54** A linear 2-port network is shown in Fig. (a). An ideal DC voltage source of 10 V is connected across Port 1. A variable resistance R is connected across Port 2. As R is varied, the measured voltage and current at Port 2 is shown in Fig. (b) as a V_2 versus $-I_2$ plot. Note that for $V_2 = 5$ V, $I_2 = 0$ mA, and for $V_2 = 4$ V, $I_2 = -4$ mA. When the variable resistance R at Port 2 is replaced by the load shown in Fig. (c), the current I_2 is _____ mA (rounded off to one decimal place).

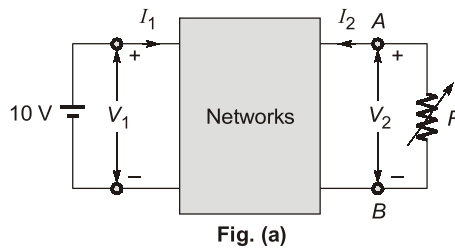


Fig. (a)

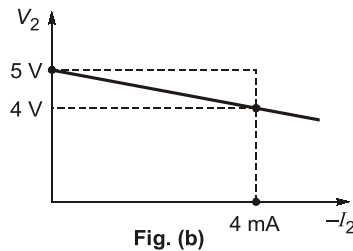


Fig. (b)

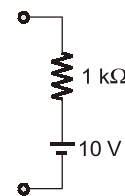
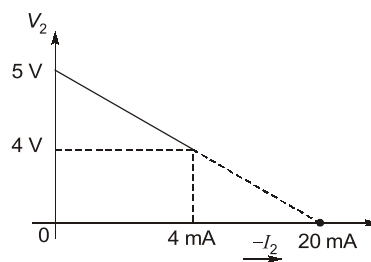


Fig. (c)

Ans. (4)

For $I_2 = 0$, $V_2 = V_{OC} = 5$ V
For Thevenin's resistance R_{th} ,



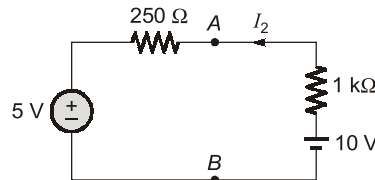
For $-I_2 = 20 \text{ mA}$, $V_2 = 0$

$$I_{SC} = -I_2$$

$$I_{SC} = 20 \text{ mA}$$

$$R_{th} = \frac{V_{OC}}{I_{SC}} = \frac{5}{20} \times 10^3 = 250 \Omega$$

Network is replaced by Thevenin's equivalent,



$$I_2 = \frac{10 - 5}{1.25 \times 10^3} = \frac{5}{1.25} \times 10^{-3} \text{ A} = 4 \text{ mA}$$

End of Solution

Q.55 For a vector $\bar{x} = [x[0], x[1], \dots, x[7]]$, the 8-point discrete Fourier transform (DFT) is denoted by $\bar{X} = \text{DFT}(\bar{x}) = [X[0], X[1], \dots, X[7]]$, where

$$X[k] = \sum_{n=0}^7 x[n] \exp\left(-j \frac{2\pi}{8} nk\right).$$

Here, $j = \sqrt{-1}$. If $\bar{x} = [1, 0, 0, 0, 2, 0, 0, 0]$ and $\bar{y} = \text{DFT}(\bar{x})$, then the value of $y[0]$ is _____ (rounded off to one decimal place).

Ans. (8)

$$y(n) = \text{DFT}[\text{DFT}(x(n))]$$

Using Duality property

$$x(n) \xrightarrow{\text{DFT}} \xrightarrow{\text{DFT}} Nx(-k) = y(n)$$

$$y(n) = Nx(-k) = Nx(N-k)$$

$$y(n) = 8(1, 0, 0, 0, 2, 0, 0, 0)$$

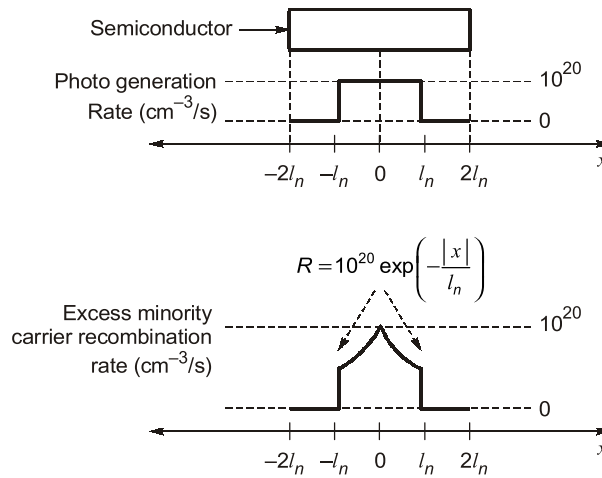
At

$$x = 0$$

$$y(0) = 8 \times 1 = 8$$

End of Solution

Q.56 A p-type semiconductor with zero electric field is under illumination (low level injection) in steady state condition. Excess minority carrier density is zero at $x = \pm 2l_n$, where $l_n = 10^{-4} \text{ cm}$ is the diffusion length of electrons. Assume electronic charge, $q = -1.6 \times 10^{-19} \text{ C}$. The profiles of photo-generation rate of carriers and the recombination rate of excess minority carriers (R) are shown. Under these conditions, the magnitude of the current density due to the photo-generated electrons at $x = +2l_n$ is _____ mA/cm^2 (rounded off to two decimal places).



Ans. (0.59)

$$\delta n(x) = R\tau_n = 10^{20} e^{-|x|/l_n} \tau_n$$

$$\therefore \delta n(l_n) = 10^{20} e^{-1} \tau_n \quad \dots(i)$$

$$l_n \leq x \leq 2l_n$$

Continuity equation in steady state,

$$D_n \frac{\partial^2 \delta n}{\partial x^2} + G - R = 0$$

$$\text{Since, } \left. \begin{array}{l} G = 0 \\ R = 0 \end{array} \right\} l_n \leq x \leq 2l_n$$

$$\therefore D_n \frac{\partial^2 \delta n}{\partial x^2} = 0$$

Whose solution is, $\delta n(x) = Ax + B$

$$\text{Since at } x = 2l_n : \delta n(2l_n) = 0 \text{ (given)}$$

$$0 = A(2l_n) + B$$

$$A = -\frac{B}{2l_n}$$

$$\therefore \delta n(x) = -\frac{B}{2l_n} x + B = B \left(1 - \frac{x}{2l_n} \right) \quad \dots(ii)$$

$$\therefore \text{At } x = l_n : \text{ equation (i) = equation (ii)}$$

$$10^{20} e^{-1} \tau_n = B \left(1 - \frac{l_n}{2l_n} \right)$$

$$\therefore B = 2 \times 10^{20} e^{-1} \tau_n$$

$$\therefore \delta n(x) = 2 \times 10^{20} e^{-1} \tau_n \left(1 - \frac{x}{2l_n} \right)$$

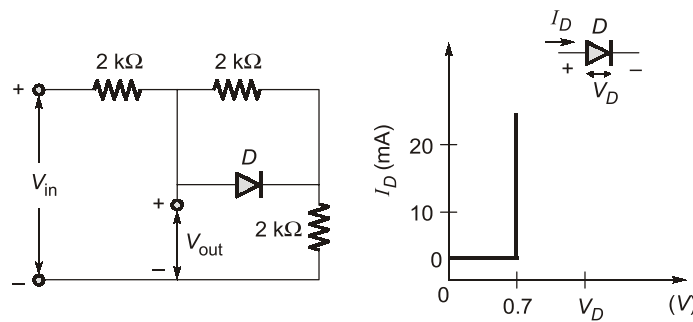
$$l_n \leq x \leq 2l_n$$

∴ Electron diffusion current density :

$$\begin{aligned} |J_n|_{\text{diff}} &= qD_n \frac{dn}{dx} = qD_n \times 2 \times 10^{20} \times e^{-1} \times \tau_n \left(0 - \frac{1}{2l_n} \right) \\ &= \frac{1.6 \times 10^{-19} \times I_n^2 \times 2 \times 10^{20} \times e^{-1}}{2l_n} \\ &= 1.6 \times 10^{-19} \times I_n \times 10^{20} \times e^{-1} \\ &= 1.6 \times 10^1 \times 1 \times 10^{-4} \times e^{-1} \quad (l_n = 10^{-4} \text{ cm}) \\ &= 0.588 \text{ mA/cm}^2 \\ &= 0.59 \end{aligned}$$

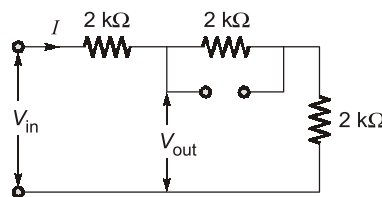
End of Solution

Q.57 A circuit and the characteristics of the diode (D) in it are shown. The ratio of the minimum to the maximum small signal voltage gain $\frac{\partial V_{\text{out}}}{\partial V_{\text{in}}}$ is _____ (rounded off to two decimal places).



Ans. (0.75)

When diode is OFF:

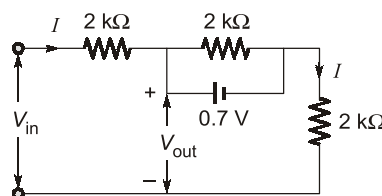


$$V_o = 4I$$

$$V_{\text{in}} = 6I$$

$$A_v = \frac{V_o}{V_{\text{in}}} = \frac{4}{6} = \frac{2}{3}$$

When diode is ON, voltage drop across diode is 0.7 V.



$$V_o = 0.7 + (2 \text{ k}\Omega)I$$

$$V_{in} = 0.7 + (4 \text{ k}\Omega)I$$

$$I = \frac{V_o - 0.7}{2 \text{ k}\Omega}$$

$$V_{in} = 0.7 + (4 \text{ k}\Omega) \cdot \frac{(V_o - 0.7)}{(2 \text{ k}\Omega)} = 0.7 + 2V_o - 1.4$$

$$V_o = \frac{V_{in} + 0.7}{2}$$

$$A'_v = \frac{\partial V_o}{\partial V_{in}} = \frac{1}{2}$$

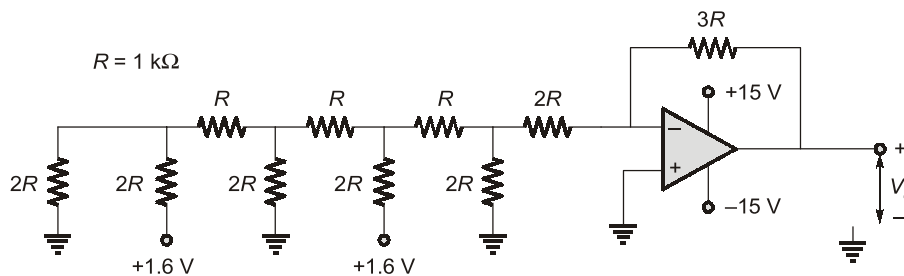
Maximum voltage gain, $A_v = \frac{2}{3}$

Minimum voltage gain, $A'_v = \frac{1}{2}$

$$\frac{(\partial V_{out} / \partial V_{in})_{\min}}{(\partial V_{out} / \partial V_{in})_{\max}} = \frac{A'_v}{A_v} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4} = 0.75$$

End of Solution

Q.58 Consider the circuit shown with an ideal OPAMP. The output voltage V_o is _____ V (rounded off to two decimal places).



Ans. **(-0.5)**

Analog output $V_o = -\text{Resolution} \times \text{Gain} \times \text{Decimal equivalent of binary data}$

$$\text{Resolution} = \frac{V_r}{2^n} = \frac{1.6}{2^4} = 0.1$$

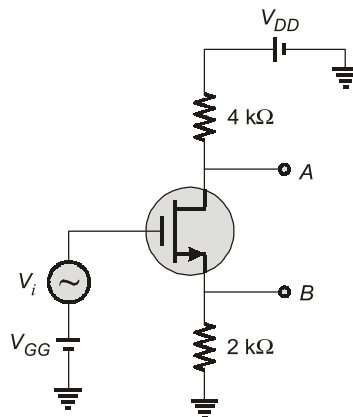
$$\text{Decimal equivalent} = 5$$

$$\text{Gain} = 1$$

$$V_o = -(0.1)(5)(1) = -0.5 \text{ V}$$

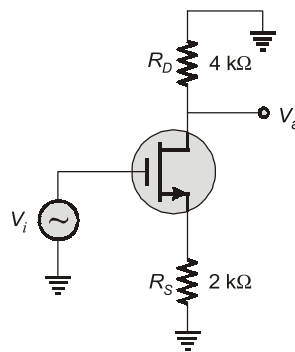
End of Solution

Q.59 Consider the circuit shown with an ideal long channel n MOSFET (enhancement mode, substrate is connected to the source). The transistor is appropriately biased in the saturation region with V_{GG} and V_{DD} such that it acts as a linear amplifier. v_i is the small-signal ac input voltage. v_A and v_B represent the small-signal voltages at the nodes A and B, respectively. The value of $\frac{v_A}{v_B}$ is _____ (rounded off to one decimal place).



Ans. (-2)

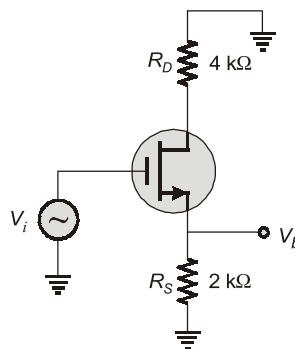
Consider V_a as an output then the small signal model is



CS in bypass amplifier

$$\text{Voltage gain } \frac{V_a}{V_{in}} = \frac{-g_m R_D}{1 + g_m R_S} \quad \dots(1)$$

Consider V_b as an output, then the small signal model,



CD amplifier

$$\text{Voltage gain } \frac{V_b}{V_{in}} = \frac{g_m R_S}{1 + g_m R_S} \quad \dots(2)$$

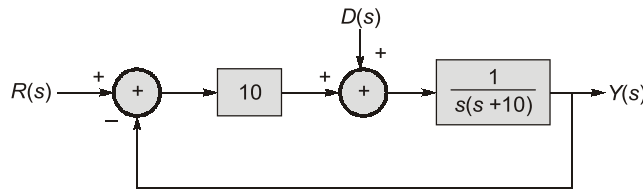
On dividing equation (1) and (2)

$$\frac{V_a/V_{in}}{V_b/V_{in}} = \frac{-R_D}{R_s}$$

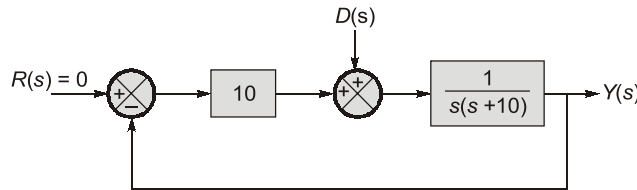
$$\frac{V_a}{V_b} = \frac{-4k}{2k} = -2$$

End of Solution

- Q.60** The block diagram of a closed-loop control system is shown in the figure. $R(s)$, $Y(s)$, and $D(s)$ are the Laplace transforms of the time-domain signals $r(t)$, $y(t)$, and $d(t)$, respectively. Let the error signal be defined as $e(t) = r(t) - y(t)$. Assuming the reference input $r(t) = 0$ for all t , the steady-state error $e(\infty)$, due to a unit step disturbance $d(t)$, is _____ (rounded off to two decimal places).



Ans. (-0.1)



$$G_1(s) = 10, \quad G_2(s) = \frac{1}{s(s+10)}$$

$$\frac{E(s)}{D(s)} = \frac{-G_2(s)}{1 + G_1(s)G_2(s)}$$

$$= \frac{-1}{s(s+10)} \cdot \frac{1}{1 + \left(10 \times \frac{1}{s(s+10)}\right)}$$

$$= \frac{-1}{s^2 + 10s + 10}$$

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s)$$

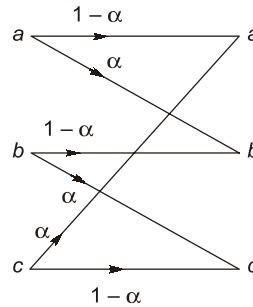
$$e_{ss} = \lim_{s \rightarrow 0} \frac{s(1/s)(-1)}{s^2 + 10s + 10}$$

$$e_{ss} = \frac{-1}{10}$$

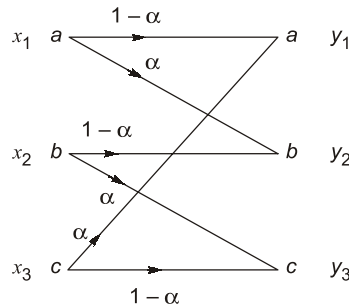
$$e_{ss} = -0.1$$

End of Solution

- Q.61** The transition diagram of a discrete memoryless channel with three input symbols and three output symbols is shown in the figure. The transition probabilities are as marked. The parameter α lies in the interval $[0.25, 1]$. The value of α for which the capacity of this channel is maximized, is _____ (rounded off to two decimal places).



Ans. (1)



Channel capacity, $C_s = \text{Max}[I(X; Y)]$

$$I(X; Y) = H(Y) - H\left(\frac{Y}{X}\right)$$

where, $H\left(\frac{Y}{X}\right) = -\sum_{i=1}^3 \sum_{j=1}^3 P(x_i, y_j) \log_2 P\left(\frac{y_j}{x_i}\right)$

$$\left[P\left(\frac{Y}{X}\right) \right] = \begin{matrix} & \begin{matrix} y_1 & y_2 & y_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 1-\alpha & \alpha & 0 \\ 0 & 1-\alpha & \alpha \\ \alpha & 0 & 1-\alpha \end{bmatrix} \end{matrix}$$

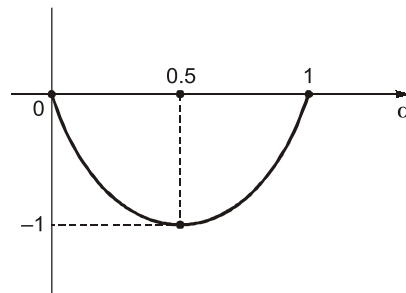
For simplification convenience, let $[P(X)] = [1 \ 0 \ 0]$

$$[P(X, Y)] = [P(X)]_a \cdot \left[P\left(\frac{Y}{X}\right) \right]$$

$$[P(X, Y)] = \begin{bmatrix} 1-\alpha & \alpha & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$H\left(\frac{Y}{X}\right) = -\{(1-\alpha)\log_2(1-\alpha) + \alpha\log_2\alpha\}$$

$$\begin{aligned}
 I(X; Y) &= H(Y) + (1 - \alpha) \log_2 (1 - \alpha) + \log_2 \alpha \\
 C_s &= \text{Max}\{I(X; Y)\} \\
 &= \text{Max}\{H(Y)\} + (1 - \alpha) \log_2 (1 - \alpha) + \alpha \log_2 \alpha \\
 C_s &= \log_2 3 + \underbrace{(1 - \alpha) \log_2 (1 - \alpha) + \alpha \log_2 \alpha}_{\alpha \log_2 \alpha + (1 - \alpha) \log_2 (1 - \alpha)}
 \end{aligned}$$



C_s will be maximum at $\alpha = 0$ and 1 .

Given $\alpha \in [0.25, 1]$

So, that $\alpha = 1$ will be the correct answer.

End of Solution

Q.62 Consider communication over a memoryless binary symmetric channel using a (7, 4) Hamming code. Each transmitted bit is received correctly with probability $(1 - \epsilon)$, and flipped with probability ϵ . For each codeword transmission, the receiver performs minimum Hamming distance decoding, and correctly decodes the message bits if and only if the channel introduces at most one bit error.

For $\epsilon = 0.1$, the probability that a transmitted codeword is decoded correctly is _____ (rounded off to two decimal places).

Ans. (0.85)

Given (7, 4) Hamming code.

Number of bits in the transmitted codeword = 7.

Given is binary symmetric channel $\rightarrow P(0/1) = P(1/0)$

$$P(0/1) = P(1/0) = \epsilon = 0.1$$

Probability of correct decoding of codeword (P_c) = Probability of atmost one bit error

$$P_c = \text{No error (or) 1 bit error}$$

When 'n' bits transmitted, probability of getting error in 'r' bits is ${}^nC_r P^r (1 - P)^{n-r}$

Where 'p' is bit error probability

$$P(1/0) = P(0/1) = 0.1$$

$$\begin{aligned}
 P_c &= {}^7C_0 (0.1)^0 (1 - 0.1)^{7-0} + {}^7C_1 (0.1)^1 (1 - 0.1)^{7-1} \\
 &= (0.9)^7 + 7 \times 0.1 \times (0.9)^6
 \end{aligned}$$

$$P_c = 0.8503$$

End of Solution

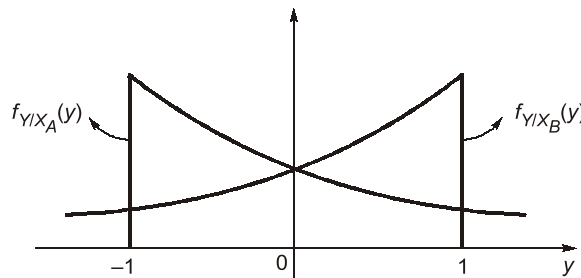
Q.63 Consider a channel over which either symbol x_A or symbol x_B is transmitted. Let the output of the channel Y be the input to a maximum likelihood (ML) detector at the receiver. The conditional probability density functions for y given x_A and x_B are:

$$f_{Y|X_A}(y) = e^{-(y+1)} u(y+1),$$

$$f_{Y|X_B}(y) = e^{(y-1)} (1 - u(y-1)),$$

where $u(\cdot)$ is the standard unit step function. The probability of symbol error for this system is _____ (rounded off to two decimal places).

Ans. (0.23)



ML decoding $\rightarrow f_{Y|X_A}(y) \underset{X_B}{\overset{X_A}{>}} f_{Y|X_B}(y)$

i.e. $f_{Y|X_A}(y) > f_{Y|X_B}(y) \rightarrow$ Decision favour of X_A

i.e. $f_{Y|X_A}(y) < f_{Y|X_B}(y) \rightarrow$ Decision favour of X_B

For $-1 < y < 0$ and $1 < y < \infty \rightarrow f_{Y|X_A}(y) > f_{Y|X_B}(y)$

For above internal decision in favour of X_A .

For $-\infty < y < -1$ and $0 < y < 1 \rightarrow f_{Y|X_B}(y) > f_{Y|X_A}(y)$

For above internal decision in favour of X_B .

$$P_e = P(X_A) \cdot P_{e_{X_A}} + P(X_B) \cdot P_{e_{X_B}}$$

$P_{e_{X_A}} \rightarrow$ Probability of error X_A transmitted

$P_{e_{X_B}} \rightarrow$ Probability of error X_B transmitted

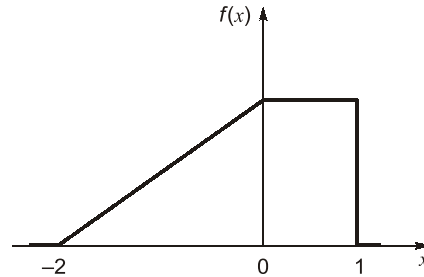
$$\begin{aligned} P_{e_{X_A}} &= \int_0^1 f_{Y|X_A}(y) dy = \int_0^1 e^{-(y+1)} u(y+1) dy \\ &= \int_0^1 e^{-(y+1)} dy = e^{-1} - e^{-2} = 0.23 \end{aligned}$$

$$\begin{aligned} P_{e_{X_B}} &= \int_{-1}^0 f_{Y|X_B}(y) dy = \int_{-1}^0 e^{(y-1)} [1 - u(y-1)] dy \\ &= \int_{-1}^0 e^{y-1} dy = e^{-1} - e^{-2} = 0.23 \end{aligned}$$

$$\begin{aligned} P_e &= P(X_A) \times 0.23 + P(X_B) \times 0.23 = 0.23[P(X_A) + P(X_B)] \\ P_e &= 0.23 \end{aligned}$$

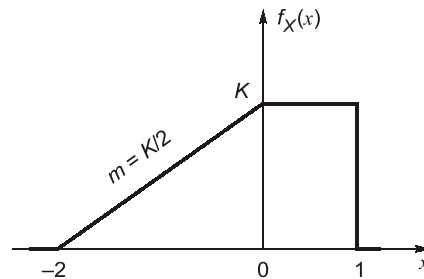
End of Solution

- Q.64** Consider a real valued source whose samples are independent and identically distributed random variables with the probability density function, $f(x)$, as shown in the figure.



Consider a 1 bit quantizer that maps positive samples to value α and others to value β . If α^* and β^* are the respective choices for α and β that minimize the mean square quantization error, then $(\alpha^* - \beta^*) =$ _____ (rounded off to two decimal places).

Ans. (1.167)



$$\frac{1}{2} \times K \times 2 + 1 \times K = 1 \Rightarrow K = \frac{1}{2} = 0.5$$

$$-2 \leq x \leq 0 \rightarrow f_X(x) = mx + C$$

$$f_X(x) = 0.25x + C$$

$$\text{when } x = -2 \rightarrow f_X(x) = 0$$

$$0 = 0.25 \times -2 + C \Rightarrow C = 0.5$$

$$f_X(x) = \frac{1}{4}x + \frac{1}{2} = -2 \leq x \leq 0$$

$$f_X(x) = 0.5 \quad ; \quad 0 \leq x \leq 1$$

Quantizer output,

$$x_q = \alpha \quad ; \quad \text{for } 0 \leq x \leq 1$$

$$x_q = \beta \quad ; \quad \text{for } -2 \leq x \leq 0$$

Mean square quantization error,

$$\text{MSQ}[Q_e] = E[Q_e^2]$$

$$Q_e = (\text{Sampled value}) - (\text{quantized value})$$

$$= X - x_q$$

$$\text{MSQ}[Q_e] = E[(X - x_q)^2]$$

$$\text{Quantization noise power, } N_o = \text{MSQ}[Q_e] = \int (X - x_q)^2 f_X(x) dx$$

$$\text{For } -2 \leq x \leq 0 \rightarrow N_Q = \int_{-2}^0 (x - \beta)^2 \left(\frac{1}{4}x + \frac{1}{2} \right) dx$$

$$= \int_{-2}^0 (x^2 + \beta^2 - 2x\beta) \left(\frac{x}{4} + \frac{1}{2} \right) dx$$

$$N_Q = \frac{\beta^2}{2} + \frac{2}{3}\beta - \frac{1}{3}$$

To find β value for which N_Q will be minimum $\rightarrow$

$$\frac{dN_Q}{d\beta} = 0 \Rightarrow \frac{1}{2} \times 2\beta + \frac{2}{3} = 0$$

$$\beta = -\frac{2}{3}$$

For $0 \leq x \leq 1 \rightarrow$

$$N_Q = \int_0^1 (x - \alpha)^2 \times \frac{1}{2} dx$$

$$N_Q = \frac{1}{6} [(1 - \alpha)^3 + \alpha^3]$$

To find ' α ' value for which N_Q is minimum

$$\frac{dN_Q}{d\alpha} = 0 \Rightarrow \frac{1}{6} [3(1 - \alpha)^2(-1) + 3\alpha^2] = 0$$

$$\alpha = \frac{1}{2}$$

Chosen α^* and β^* for which mean square quantization error (N_Q) is minimum will be

$\frac{1}{2}$ and $-\frac{2}{3}$ respectively.

$$\alpha^* - \beta^* = \frac{1}{2} + \frac{2}{3} = \frac{7}{6} = 1.167$$

Q.65 In an electrostatic field, the electric displacement density vector, $\vec{D}$, is given by

$$\vec{D}(x, y, z) = (x^3 \vec{i} + y^3 \vec{j} + xy^2 \vec{k}) \text{ C/m}^2,$$

where $\vec{i}, \vec{j}, \vec{k}$ are the unit vectors along x -axis, y -axis, and z -axis, respectively. Consider a cubical region R centered at the origin with each side of length 1 m, and vertices at $(\pm 0.5 \text{ m}, \pm 0.5 \text{ m}, \pm 0.5 \text{ m})$. The electric charge enclosed within R is _____ C (rounded off to two decimal places).

Ans. (0.5)

$$Q_{\text{enclosed}} = \int_V \rho_v dv = \int (\nabla \cdot \vec{D}) dv$$

$$\nabla \cdot \vec{D} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(xy^2)$$

$$\nabla \cdot \vec{D} = (3x^2 + 3y^2) = 3(x^2 + y^2)$$

$$dv = dx dy dz$$

$$\begin{aligned}
 Q_{\text{enc}} &= \int_V 3(x^2 + y^2) dx dy dz \\
 &= 3 \int_{x=-0.5}^{0.5} \int_{y=-0.5}^{0.5} \int_{z=-0.5}^{0.5} (x^2 + y^2) dx dy dz \\
 &= 3 \left[\left\{ \frac{x^3}{3} \right\}_{-0.5}^{0.5} \{y\}_{-0.5}^{0.5} \{z\}_{-0.5}^{0.5} + \{x\}_{-0.5}^{0.5} \left\{ \frac{y^3}{3} \right\}_{-0.5}^{0.5} \{z\}_{-0.5}^{0.5} \right] \\
 &= \left((0.5)^3 - (-0.5)^3 \right) (0.5 + 0.5) (0.5 + 0.5) \times 2 \\
 Q_{\text{enc}} &= 0.5 \text{ C}
 \end{aligned}$$

End of Solution

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